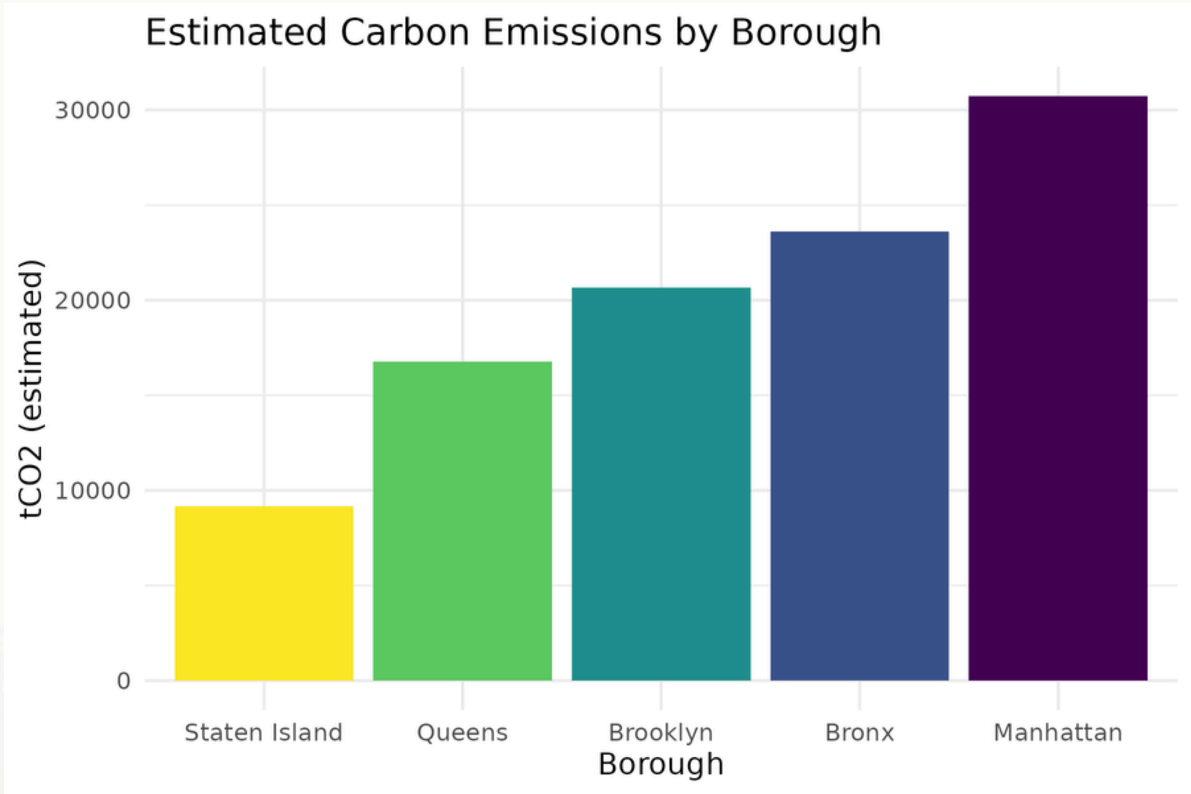
Energy, Emissions, and Equity in NYC Public Schools

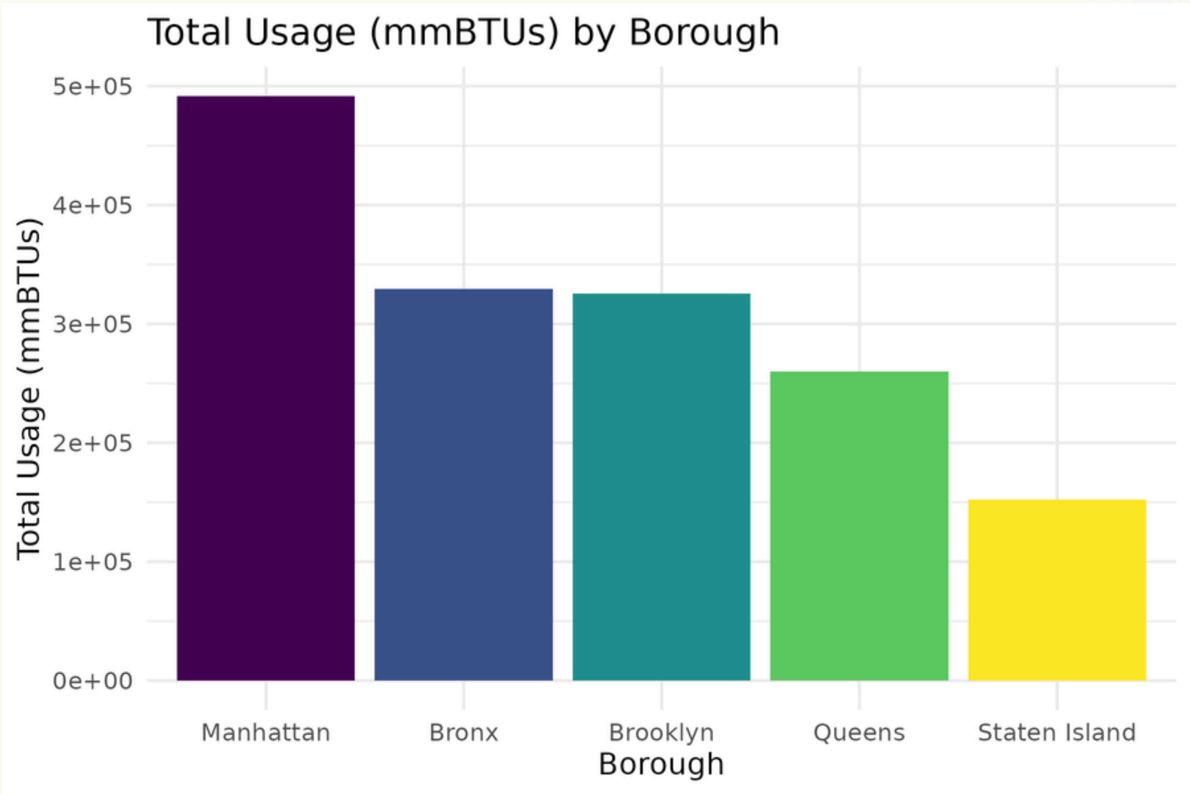
Research Question:

How does energy consumption vary across NYC public school buildings, how does this relate to student enrollment and socioeconomic conditions, and what does this imply for environmental equity and infrastructure planning?

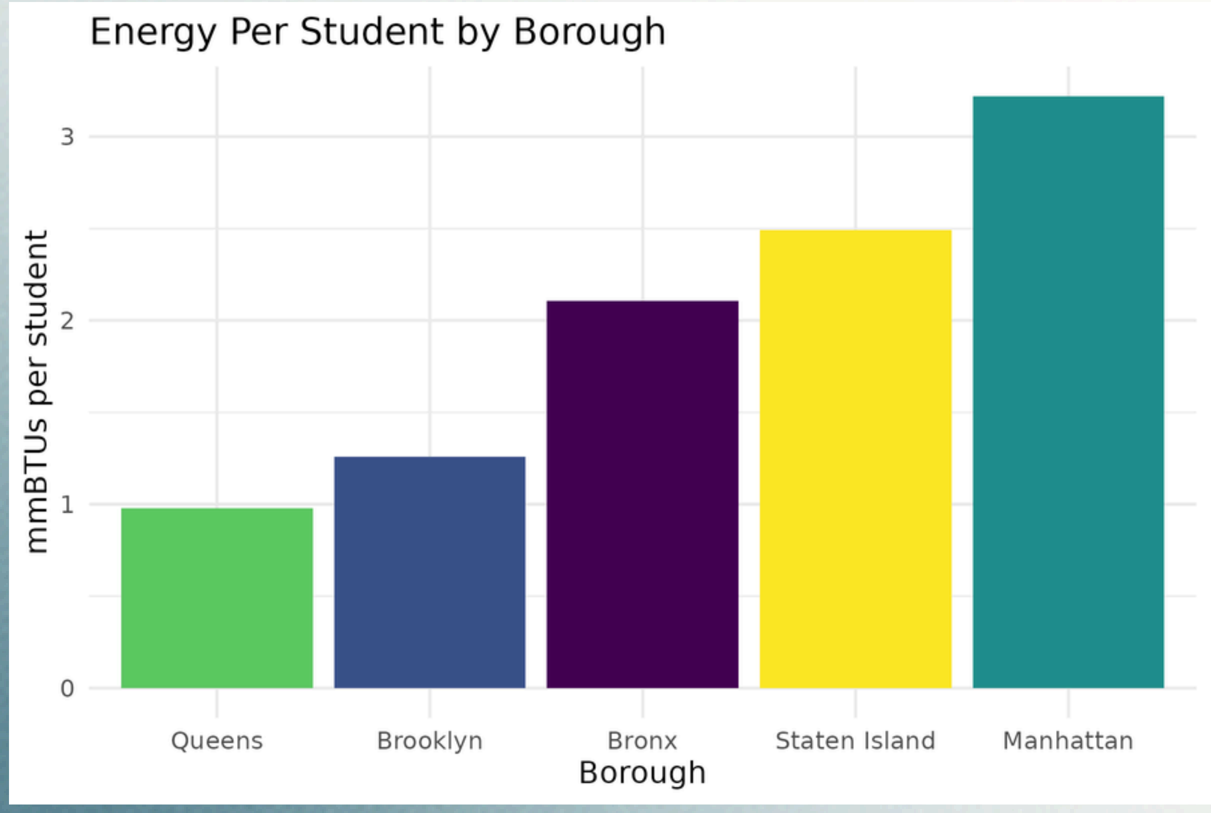
The dataset includes:

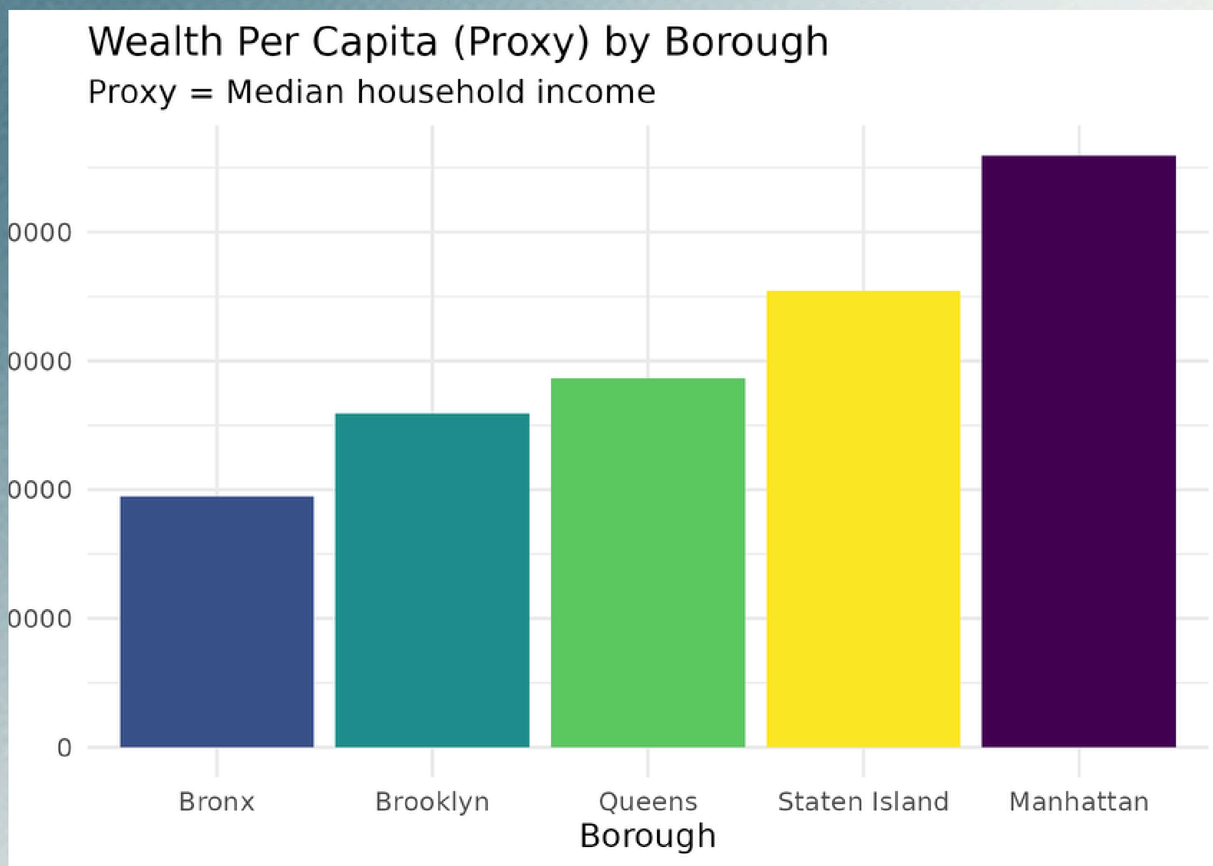
- Borough-level energy consumption from NYC building-level usage data
- Enrollment from NYSED's 2024–25 "All Students" district-level file
- Socioeconomic conditions from the Neighborhood Financial Health dataset
- Derived carbon emissions using EPA conversion factors





Noting the carbon chart's log scale—the pattern is: wealthier boroughs/schools consume more energy resources yet show lower emissions intensity (emissions rise less than energy), consistent with cleaner fuels and newer systems. In contrast, lower-income boroughs (especially the Bronx) have higher energy per student and higher emissions intensity, pointing to older, less efficient infrastructure.



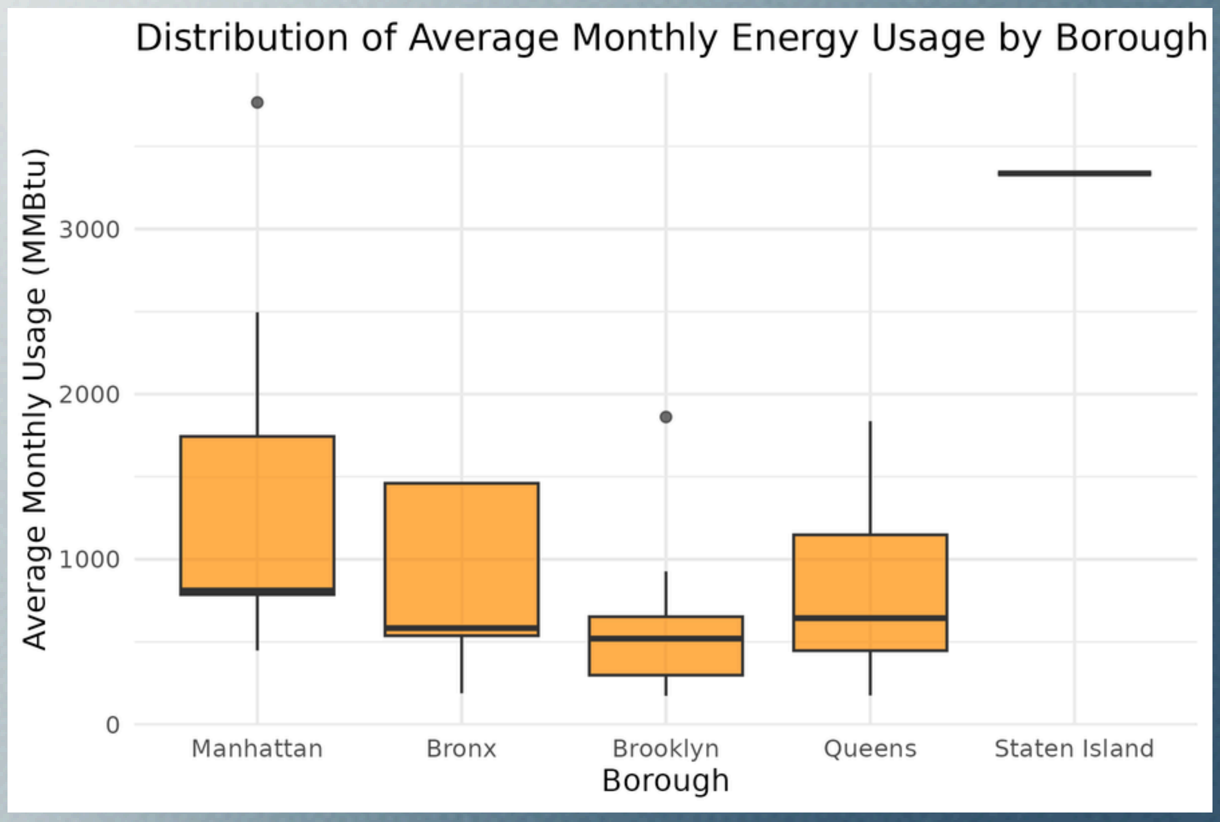


The Total Energy Use by Borough graph shows that Manhattan leads by a wide margin, followed by the Bronx and Brooklyn, with Staten Island using the least. This reflects where the largest share of energy resources is directed within the school system.

The Carbon Emissions by Borough graph mirrors that pattern because NYC schools rely on similar fossil-fuel energy sources. Boroughs receiving more total energy also produce more emissions.

Put together, the two graphs show that boroughs with wealthier student populations—especially Manhattan—receive more total energy, while lower-income boroughs have access to far fewer energy resources. The environmental burden aligns with this distribution: the areas receiving the most energy also create the highest carbon footprint.

1 mmBTU = 8 gallons gas



The box plot shows clear infrastructure differences: Manhattan and Staten Island have tight, stable monthly usage, while the Bronx has the widest spread and largest spikes. This instability reflects older, inefficient buildings that require inconsistent energy input and once again that energy inefficiency is concentrated in lower-income boroughs.

Note: The energy data does not separate fuel types or account for building renovations, and student counts are aggregated at the district level, so per-student estimates assume uniform distribution across buildings. However, these limitations do not undermine the core patterns; they frame the conclusions appropriately.